

```
import re
import string
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.tokenize import TweetTokenizer
```

```
tweet1= tweets["Tweets"][2005]
tweet1
```

```
'@TnmntTiles Our tiles, while a nice green, are boring - but I love the original glass
door the guy downstairs has :) http://t.co/fxgpk8hNiC'
```

```
tweet2 = re.sub(r'^RT[\s]+', '', tweet1)

tweet2 = re.sub(r'https?:\/\/[^\s\n\r]+', '', tweet2)

tweet2 = re.sub(r'#', '', tweet2)
tweet2 = re.sub(r'@\w+', '', tweet2)
tweet2 = re.sub(r'[:]|:()|', '', tweet2)
print(tweet1)
print("\n Cleaned tweet: \n",tweet2)
```

```
@TnmntTiles Our tiles, while a nice green, are boring - but I love the original glass c

Cleaned tweet:
Our tiles, while a nice green, are boring - but I love the original glass door the gu
```

```
tokenizer = TweetTokenizer(preserve_case=False, strip_handles=T

tweet_tokens = tokenizer.tokenize(tweet2)

print()
print("Tokenized string:\n")
print(tweet_tokens)
```

```
Tokenized string:
```

```
['our', 'tiles', ',', 'while', 'a', 'nice', 'green', ',', 'are', 'boring', '-', 'but',
```

```
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
True
```

```
stopwords_english = stopwords.words('english')
print('stop words\n')
print(stopwords_english)
```

```
print('\npunctuation\n')
print(string.punctuation)
```

stop words

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'and', 'as', 'at', 'be', 'because', 'before', 'big', 'but', 'by', 'can', 'could', 'did', 'do', 'down', 'each', 'few', 'for', 'from', 'further', 'had', 'has', 'have', 'he', 'her', 'his', 'how', 'i', 'if', 'in', 'into', 'is', 'it', 'its', 'me', 'more', 'most', 'my', 'no', 'nor', 'not', 'of', 'off', 'on', 'once', 'only', 'or', 'other', 'out', 'over', 'so', 'some', 'than', 'that', 'the', 'their', 'them', 'there', 'these', 'they', 'this', 'those', 'through', 'to', 'too', 'under', 'until', 'up', 'us', 'very', 'was', 'we', 'were', 'what', 'when', 'where', 'which', 'while', 'who', 'whom', 'why', 'with', 'without', 'you', 'your']
```

punctuation

```
!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~
```

```
print(tweet_tokens)
print()
tewwts_clean = []
for word in tweet_tokens:
    if (word not in stopwords_english and word not in string.punctuation):
        tewwts_clean.append(word)
print("removed stop words and punctuation:")
print(tewwts_clean)
```

```
['our', 'tiles', ',', 'while', 'a', 'nice', 'green', ',', 'are', 'boring', '-', 'but', 'down', 'the', 'door', 'guy', 'owns', 'a', 'house']
```

removed stop words and punctuation:

```
['tiles', 'nice', 'green', 'boring', 'love', 'original', 'glass', 'door', 'guy', 'downs', 'house']
```

```
def process_tweets(tweet):
    tokenizer = TweetTokenizer(preserve_case=False, strip_handles=True,
                               reduce_to_ngrams=True)

    tweet = re.sub(r'^RT[\s]+', '', tweet)
    tweet = re.sub(r'https?:\/\/[^\s\n\r]+', '', tweet)
    tweet = re.sub(r'#', '', tweet)

    tweet_tokens = tokenizer.tokenize(tweet)
    stopwords_english = stopwords.words('english')
    tweets_clean = []
    for word in tweet_tokens:
        if (word not in stopwords_english and word not in string.punctuation):
            tweets_clean.append(word)
    return tweets_clean
```

```
import nltk
import pandas as pd
import re
import string
from nltk.corpus import twitter_samples, stopwords
from nltk.tokenize import TweetTokenizer
```

```
nltk.download('twitter_samples')
```

```

nltk.download('stopwords')

def process_tweets(tweet):
    tokenizer = TweetTokenizer(preserve_case=False, strip_hand

    tweet = re.sub(r'^RT[\s]+', '', tweet)
    tweet = re.sub(r'https?:\/\/[^\s\n\r]+', '', tweet)
    tweet = re.sub(r'#', '', tweet)

    tweet_tokens = tokenizer.tokenize(tweet)
    stopwords_english = stopwords.words('english')
    tweets_clean = []
    for word in tweet_tokens:
        if (word not in stopwords_english and word not in stri
            tweets_clean.append(word)
    return tweets_clean

tweets = twitter_samples.strings()
all_tweets = pd.DataFrame(tweets, columns=['Tweets'])

processed_tweet = all_tweets['Tweets'].apply(process_tweets)
all_tweets['processed_tweets'] = processed_tweet
pd.set_option('display.max_colwidth', None)
all_tweets

```

```
[nltk_data] Downloading package twitter_samples to /root/nltk_data...
[nltk_data] Package twitter_samples is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

	Tweets	processed_tweets
0	hopeless for tmr :(	[hopeless, tmr, :(]
1	Everything in the kids section of IKEA is so cute. Shame I'm nearly 19 in 2 months :(	[everything, kids, section, ikea, cute, shame, nearly, 19, 2, months, :(]
2	@Hegelbon That heart sliding into the waste basket. :(	[heart, sliding, waste, basket, :(]
3	"@ketchBurning: I hate Japanese call him "bani" :(: ("n\nMe too	[" , hate, japanese, call, bani, :(, :(, "]
4	Dang starting next week I have "work" :(	[dang, starting, next, week, work, :(]
...	...	...
29995	RT @UKLabour: .@Ed_Miliband: we're not going to do a deal with the SNP, there will be no coalition with the SNP. #bbcqt\nhttps://t.co/n65hTe...	[going, deal, snp, coalition, snp, bbcqt]
29996	RT @DisabledScot: @blairmcdougall @ScotlandTonight @bbcqt If Ed thinks Scots will suddenly rush to vote Labour, we'll vote SNP, Labour just...	[ed, thinks, scots, suddenly, rush, vote, labour, vote, snp, labour, ...]
29997	RT @Staircase2: @VividRicky exactly but that alleged commute was made up by Farage as was his 'discomfort' in allegedly 'not hearing Englis...	[exactly, alleged, commute, made, farage, discomfort, allegedly, hearing, englis, ...]
29998	Actually agreed with %95 of what farage was sayin rhen#voteukip	[actually, agreed, 95, farage, sayin, rhenvoteukip]
29999	RT @PeatWorrier: "A vote for the SNP is. Er. A vote for another referendum we insist Holyrood has no legal authority to legislate for" http...	[vote, snp, er, vote, another, referendum, insist, holyrood, legal, authority, legislate, http, ...]

30000 rows × 2 columns

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```
# Import required libraries
import re
import string
import nltk
!pip install emoji
import emoji

# Download required NLTK resources
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
```